

PAH-OMC-016: radial tightness and uniform matter nondegeneracy

TECT verification-first repository
claim C6-SPACETIME-SIGNATURE

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Purpose and scope

This checkpoint tests the single preregistered PAH-OMC-016 candidate. The functional, rates, mobility, projection and ordered limits are unchanged. The preregistration is PAH-OMC-016-resolved-radial-prereg-v1.json, SHA-256 (the line break is typographic only):

```
1cebe3acff477175125c7abf2ebdfa2cd5  
b65089530ae3581bbaa69b23c161b7
```

The source crosswalk and a second proof presentation are in PAH-OMC-016-uniform-certificate.md. All six immutable parent hashes are checked by both executable reconstructions.

Use exactly the finite relational strip G_n of OMC-004, $n \geq 2$, including its unsplit frontier square and boundary interactions. Write $N = 2(n + 2)$, $E = 4n + 4$, $P = 2n + 1$. Its degree is at most five, directly from the two horizontal, one vertical and at most two diagonal incidences. No physical dimension, length or volume is assigned to the strip. Set $K = 2$, $M_s = 1$, $\epsilon = 1/2$, $\beta = \nu = 1$, $m^2 = \theta = 0$, and all other displayed couplings equal to one. For $j \geq 0$,

$$R_j = 2^j, \quad M_j = 2^{2j}, \quad h_j = 2^{-j}, \quad r_v = h_j \ell_v, \quad 0 \leq \ell_v \leq M_j.$$

Let z contain every aperture, phase and representative link label. Its cardinality is 2^{2N+E} ; phases at zero amplitude remain distinct. The deterministic grade $Q = \sum_v \ell_v$ adds no multiplicity. The declared weights $w_Q = Z_Q / \sum_Q Z_Q$ therefore give

$$\mu_{n,j}(x) = Z_{n,j}^{-1} e^{-F(x)}$$

on the full labelled counting domain, without a radial Jacobian or probe. The witnesses are exactly $b_v = \min(1, r_v)$, $v = a, d$. They are not the old raw or normalized integer labels for $j > 0$.

Statement. At each fixed n , these laws are tight uniformly in j and converge weakly to the finite-graph radial Gibbs law ν_n below. For every vertex,

$$\lim_{j \rightarrow \infty} \mu_{n,j}(b_v^2) = \nu_n(b_v^2) \geq c := \frac{11}{80} e^{-1550/3} > 0, \quad n \geq 2.$$

The same c works for all n ; no limit in n is taken here. It is a conservative derived lower bound, not a fitted expectation.

Exact energy and fixed-graph cutoff limit

Write $s_v \in \{1/2, 1\}$ and let $\sigma_e \in \{-1, 1\}$ be the product of the endpoint phase and link signs. The source formula is exactly

$$F(r, z) = F_0(z) + \sum_v \left(\frac{r_v^6}{6} + \frac{r_v^4}{4} + \frac{s_v^2 r_v^2}{2} \right) + \sum_{e=\{v,w\}} \frac{J_e(s)}{2} (r_v - \sigma_e r_w)^2, \quad 1 \leq J_e = \frac{2}{s_v + s_w} \leq 2.$$

Here F_0 is the unchanged nonnegative aperture/Wilson energy. Thus $F \geq \sum_v r_v^6/6$ with no term discarded from the model.

For $x \in \mathbb{R}_+^N$ define $a_j(x)_v = h_j \lfloor x_v/h_j \rfloor$, $D_j = [0, R_j + h_j)^N$ and $q_j(x, z) = \mathbf{1}_{D_j}(x) e^{-F(a_j(x), z)}$. The upper endpoint $\ell = M_j$ has its own half-open cell. Consequently

$$\sum_z \int q_j(x, z) dx = h_j^N Z_{n,j}$$

exactly. This common factor cancels from expectations; it does not change the finite reference measure. Pointwise $a_j(x) \rightarrow x$ and D_j exhausts the orthant. Since $h_j \leq 1$,

$$q_j(x, z) \leq W(x) := \prod_v w(x_v), \quad w(t) = \exp \left(-\frac{\max(t-1, 0)^6}{6} \right).$$

For $t \geq 4$, $(t-1)^6/6 \geq t$, so $\int_0^\infty w(t) dt \leq 4 + e^{-4} \leq 5$. Every polynomial moment of W is finite. Dominated convergence gives the finite, positive normalization

$$Z_n^{\text{rad}} = \sum_z \int_{\mathbb{R}_+^N} e^{-F(r, z)} dr, \quad d\nu_n = (Z_n^{\text{rad}})^{-1} e^{-F(r, z)} dr$$

with labelled counting in z . Applying the same argument to $g(a_j(x), z) q_j(x, z)$ proves weak convergence for every bounded continuous g , including b_v^2 .

An explicit all- j tightness bound uses the unit cube, where

$$F \leq C_n := \frac{25}{24}N + \frac{33}{8}E + 4P.$$

The vertex contribution is $1/8 + 1/6 + 1/4 + 1/2$, the edge contribution is $1/8 + 4$, and each Wilson term is at most $2 \cdot 2$. Every D_j contains this cube, so the lifted partition is at least $2^{2N+E} e^{-C_n}$. For $L \geq 4$, a finite union bound and $a_j(x) \leq x$ give

$$\sup_j \mu_{n,j} \{ \max_v r_v \geq L \} \leq e^{C_n} N 5^{N-1} e^{-L} \rightarrow 0.$$

This estimate is only fixed- n ; its volume-dependent constant is never used to infer the uniform lower bound.

Boundary-valid identity and a uniform local moment

All polynomial moments under ν_n are finite. For fixed other coordinates and labels, $r_v e^{-F}$ vanishes at 0 and at infinity. Its derivative, and its integral over the other coordinates, are absolutely integrable by polynomial-times-sextic domination. Integration by parts and Fubini therefore yield

$$1 = \nu_n(r_v \partial_v F) = \nu_n \left(r_v^6 + r_v^4 + s_v^2 r_v^2 + \sum_{w \sim v} J_{vw} (r_v^2 - \sigma_{vw} r_v r_w) \right).$$

The value 1 uses reference dr , not $r dr$. This identity is applied to the constructed continuous radial law, not asserted at a finite mesh.

Let $X = \max_v \nu_n(r_v^2)$, finite and attained. At a maximizing vertex, Jensen, the identity above and Cauchy–Schwarz imply

$$X^3 \leq \nu_n(r_v^6) \leq 1 + 2 \sum_{w \sim v} \nu_n(r_v r_w) \leq 1 + 10X.$$

If $X \geq 4$, then $X^3 - 10X = X(X^2 - 10) \geq 24 > 1$, a contradiction. Hence $\nu_n(r_v^2) \leq 4$ for every v, n . This is a local maximum argument, not a volume sum; it assumes no translation symmetry and retains all frontier terms.

Uniform conditional probability away from zero

Let $A_v = \{r_w \leq 8 \text{ for every } w \sim v\}$. Markov’s inequality and the degree bound give $\nu_n(A_v) \geq 1 - 5 \cdot 4/64 = 11/16$. Conditioning on all other amplitudes and z leaves density proportional to $e^{-V(t)} dt$, where

$$V(t) = \frac{t^6}{6} + \frac{t^4}{4} + \frac{At^2}{2} - Ht, \quad A = s_v^2 + \sum_{w \sim v} J_{vw} \in [1/4, 11], \quad H = \sum_{w \sim v} J_{vw} \sigma_{vw} r_w.$$

On A_v , $|H| \leq 80$. For $1 \leq t \leq 2$,

$$V(t) \leq \frac{64}{6} + \frac{16}{4} + 22 + 160 = \frac{590}{3}.$$

For $0 \leq t \leq 4$, $V(t) \geq -320$; for $t \geq 4$, $80t \leq t^6/12$ and $t \leq t^6/12$, so $V(t) \geq t$. The conditional normalization is at most $4e^{320} + e^{-4} \leq 5e^{320}$. Therefore, on A_v ,

$$\Pr_{\nu_n}\{1 \leq r_v \leq 2 \mid r_{-v}, z\} \geq \frac{1}{5} e^{-1550/3}.$$

Since $b_v^2 = 1$ on this interval, averaging gives the stated $\nu_n(b_v^2) \geq (11/80)e^{-1550/3}$. The thresholds 4, 8 are auxiliary proof constants, not model or witness changes. This conditioning is ordinary disintegration, not a new dynamics or a refinement averaging map.

Verification and literature applicability

The primary lane differentiates the exact signed-edge polynomial and derives the constants from source inputs. The independent lane directly expands coefficients using rational arithmetic, verifies all permitted signed/aperture star types, and independently derives the barriers. The certificate also gives a finite-box Riemann/tail proof: uniform continuity controls each fixed box, the integrable majorant controls its complement, and the positive partition lower bound permits normalization. Neither lane extrapolates a finite table to prove a limit.

The standard dominated convergence theorem is imported only with its measurability, pointwise convergence and integrable-majorant hypotheses verified above. A primary reference is MIT 18.102, Spring 2021, Lecture 12, Theorem 125, printed page 61; its exact URL and the finite-product/Fubini crosswalk are in the certificate. Jensen, Cauchy–Schwarz and Markov use the explicitly proved moments. No Q3LOCK or infinite-volume Gibbs theorem is imported. R-508 retains its different, restricted-cutoff negative conclusion.

Lean cross-checks signed-edge algebra, the cubic barrier, the neighbor probability arithmetic, tail inequalities, exponent assembly and strict positivity. The measure construction, integration by parts and source-to-polynomial translation remain explicit analytic proofs, not claims of a fully formalized measure-theoretic theorem. The result manifest lists the exact source, toolchain and declaration hashes. All independent and hostile reviews here are internal; no external signed referee report is claimed.

Devil’s-advocate audit

1. Wrong sign or missing edge factor $1/2$: DISMISSED within the pinned scope by independent expansion and hostile mutations.
2. A hidden polar Jacobian or lost endpoint cell: DISMISSED by the labelled cell identity and the boundary-valid dr virial formula.
3. Volume-dependent domination proves volume uniformity: UPHELD as an objection to that inference. The independent local maximum argument, not C_n , supplies the uniform estimate.
4. Negative conditional field H invalidates the small-ball bound: DISMISSED by the two-sided bound $|H| \leq 80$.
5. Positive finite weights or the new witnesses repair R-488/R-508: UPHELD against such a claim; neither old statement is changed.
6. A common positive bound proves a unique outer state or dynamics: UPHELD against that extension. Outer convergence and stationarity passage remain separate questions; no Markov-time reinterpretation is made.

Result footer

Result ID: R-509
Claim ID: C6-SPACETIME-SIGNATURE
Precise statement: Fixed- n radial weak convergence/tightness
and a common positive lower bound on both declared b_v^2 .
Verdict: PASS within the frozen PAH-OMC-016 state test.
Scope: G_n , $n \geq 2$; resolved radial j limit first;
 $K=2$, $M_s=1$, $\epsilon=1/2$, $\beta=\nu=1$; no n limit.
Dependencies: Six preregistered immutable PAH parent files.
Evidence grade: ANALYTIC, EXACT, EXECUTED.
Reproduction command: `python -X utf8`
`verification/scripts/pah_omc016_uniform_verify.py`
Expected output: All lanes PASS; diagnostics-free Lean.
Falsification gate: Source/measure/cell/energy mismatch,
invalid boundary passage, or failure of the uniform bound.
Tier before / after: T1 / T1 (host claim unchanged).
No-overclaim statement: No PAH-OMC-014 closure,
infinite-volume dynamics, continuum, physical Pre-A,
spacetime, QFT, gravity, Yang-Mills, mass gap or TOE.
Next required action: A separate bounded-cylinder Cauchy
question for the cutoff-limit laws ν_n as n grows.